

Experiment 2: Parallel Bubble Sort using OpenMP

This guide explains how to run Parallel Bubble Sort using OpenMP on both Windows and Linux/Ubuntu systems. The guide is beginner friendly and includes correct code, execution steps, output, and viva questions.

Problem Statement: Write a program to implement Parallel Bubble Sort using OpenMP and measure execution time.

Correct Program Code

```
#include <iostream>
#include <omp.h>

using namespace std;

// Function to swap two numbers
void swapNumbers(int &a, int &b)
{
    int temp = a;
    a = b;
    b = temp;
}

// Parallel Bubble Sort Function
void parallelBubbleSort(int arr[], int n)
{
    for (int i = 0; i < n; i++)
    {
        int first = i % 2;

        #pragma omp parallel for
        for (int j = first; j < n - 1; j += 2)
        {
            if (arr[j] > arr[j + 1])
            {
                swapNumbers(arr[j], arr[j + 1]);
            }
        }
    }
}

int main()
{
    int n;

    cout << "Enter total number of elements: ";
    cin >> n;

    int *arr = new int[n];

    cout << "Enter elements:\n";

    for (int i = 0; i < n; i++)
    {
        cin >> arr[i];
    }

    double start_time = omp_get_wtime();

    parallelBubbleSort(arr, n);

    double end_time = omp_get_wtime();

    cout << "\nSorted Array:\n";

    for (int i = 0; i < n; i++)
    {
        cout << arr[i] << " ";
    }

    cout << "\n\nExecution Time: "
```

```

        << (end_time - start_time) * 1000
        << " milliseconds\n";

    delete[] arr;

    return 0;
}

```

How to Run on Windows

1. Install Code::Blocks with MinGW compiler.
2. Open Code::Blocks.
3. Go to File → New → Empty File.
4. Paste the program code.
5. Save file as bubble_sort.cpp.
6. Go to Settings → Compiler.
7. Open Compiler Settings → Other Compiler Options.
8. Add: -fopenmp
9. Click OK.
10. Press F9 or click Build and Run.

How to Run on Linux / Ubuntu

Install required software:

```

sudo apt update
sudo apt install g++
sudo apt install libomp-dev

```

Compile and run:

```

g++ -fopenmp bubble_sort.cpp -o bubble_sort

./bubble_sort

```

Sample Input

```

Enter total number of elements: 5

Enter elements:
3
6
5
9
10

```

Expected Output

```

Sorted Array:
3 5 6 9 10

Execution Time: 42 milliseconds

```

Important Viva Questions

Question	Answer
Why use OpenMP?	To execute sorting operations in parallel using multiple threads.
Which OpenMP directive is used?	#pragma omp parallel for
Which sorting technique is used?	Odd-Even Bubble Sort.
What is the time complexity?	$O(n^2)$